
Enterprise WordPress Performance Operational Checklist

Version 1.0 — DRAFT

Dan Knauss, General Editor

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A checklist-oriented runbook based on the WP VIP Learn Enterprise WordPress Performance course. Use it to diagnose, optimize, and verify performance on high-traffic WordPress sites where backend execution, cacheability, database access, object caching, traffic events, and measurement discipline matter.

Do not start by installing another performance plugin. Start by identifying where time is spent, whether WordPress is executing at all, and whether the response can safely be cached.

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0. Define the performance problem

- ☐ Identify the affected surface:
 - ☐ public anonymous page
 - ☐ logged-in page
 - ☐ editorial/admin workflow
 - ☐ API endpoint
 - ☐ AJAX endpoint
 - ☐ search results
 - ☐ sitemap generation
 - ☐ redirect path
 - ☐ high-traffic event page
- ☐ Identify the failure mode:
 - ☐ high TTFB
 - ☐ slow PHP execution
 - ☐ slow MySQL queries

- ☐ too many database queries
 - ☐ cache misses
 - ☐ object-cache pressure
 - ☐ cache stampede/race condition
 - ☐ slow third-party/external request
 - ☐ client-side JavaScript delay
 - ☐ load-test failure
- ☐ Confirm the test target URL or route.
 - ☐ Record whether the request is anonymous or authenticated.
 - ☐ Record whether the page should be cacheable.
 - ☐ Confirm rollback path before changing production behavior.
-

1. Trace the request cycle

For each target request, walk the full path.

- ☐ User enters URL.
- ☐ DNS resolves the hostname.
- ☐ Browser sends HTTP request.
- ☐ Edge/CDN checks for cached response.
- ☐ Origin web server receives request if edge misses or bypasses.
- ☐ WordPress core loads.
- ☐ MU plugins load.
- ☐ Plugins and theme code run.
- ☐ Database and object cache are queried.
- ☐ Template is selected and rendered.
- ☐ HTTP response is sent to browser.
- ☐ Browser parses HTML, downloads assets, executes JavaScript, renders the page.
- ☐ Additional requests fire: AJAX, REST, media, tracking, personalization.

Decision point:

- ☐ If the edge cache returns a valid HIT, focus on payload, browser rendering, and post-load requests.
 - ☐ If the request reaches origin, focus on cacheability, PHP, database, object cache, hooks, templates, and external calls.
-

2. Establish a baseline

Run the same measurements before and after every change.

- ☐ Measure uncached TTFB.
- ☐ Measure warm-cache TTFB.
- ☐ Capture cache headers.
- ☐ Capture response status and redirects.
- ☐ Capture PHP/application timing.
- ☐ Capture database query count and slow queries.
- ☐ Capture object-cache behavior when available.
- ☐ Capture frontend impact if browser rendering is part of the symptom.

Useful command:

```
curl -s -o /dev/null -D headers.txt \
-w "dns:%{time_namelookup} connect:%{time_connect} tls:%{time_appconnect} ttfb:%{time_total}" \
https://example.com/target/
```

Baseline table:

Target	User state	Expected		TTFB	PHP time	DB time	Notes
		cacheability	Cache status				
/	anonymous	uncacheable					
/search/	anonymous	maybe	dynamic				
/wp-json/...	varies	endpoint-specific					

3. DNS and HTTP request setup

- ☐ Confirm DNS resolves quickly and predictably.
- ☐ Avoid unnecessary DNS indirection and stale records.
- ☐ Confirm final canonical URL does not require multiple redirects.
- ☐ Confirm HTTPS works without certificate or protocol negotiation issues.
- ☐ Confirm HTTP/2 or HTTP/3 support where appropriate.

- ☐ Check request headers that may vary cache behavior:
 - ☐ cookies
 - ☐ authorization
 - ☐ query strings
 - ☐ user-agent variation
 - ☐ language/country headers

Redirect check:

```
curl -IL https://example.com/some-url/
```

- ☐ If multiple redirects occur, identify the source: WordPress, plugin, CDN, web server, localization, or application code.
- ☐ Look for `x-redirect-by` when WordPress generates redirects.

Examples seen in the course:

```
x-redirect-by: WordPress
x-redirect-by: Polylang Pro
```

Debug filter example:

```
add_filter( 'x_redirect_by', function( $x_redirect_by, $status, $location ) {
    wp_die( var_dump( array(
        $x_redirect_by,
        $status,
        $location,
        wp_debug_backtrace_summary(),
    ) ) );
}, 10, 3 );
```

Use that only in safe debugging contexts, not as a production-facing change.

4. Full-page cache and edge cache

Full-page caching is the first major enterprise performance boundary: if the edge can serve HTML, WordPress and MySQL may not need to run.

- ☐ Identify pages that should be edge-cacheable.
- ☐ Identify pages that must bypass cache:

- ☐ logged-in personalized views
- ☐ carts
- ☐ checkout
- ☐ account pages
- ☐ previews
- ☐ nonce-sensitive forms
- ☐ private API responses
- ☐ Check cache status response headers.
- ☐ Confirm Cache-Control behavior is intentional.
- ☐ Confirm cookies are not causing broad cache bypasses.
- ☐ Confirm query parameters are not fragmenting the cache key unnecessarily.
- ☐ Confirm TTL is appropriate for content volatility.
- ☐ Confirm invalidation/purge logic is precise enough.
- ☐ Investigate cache misses before tuning PHP.

Edge-cache triage:

- ☐ Is the response cacheable?
 - ☐ Does it include Set-Cookie?
 - ☐ Does it vary by cookie or authorization?
 - ☐ Is the TTL too short?
 - ☐ Is a plugin sending no-cache headers?
 - ☐ Is the request URL unique because of query strings?
 - ☐ Are purges too broad or too frequent?
-

5. TTL, headers, and invalidation

- ☐ Set long TTLs for stable public content.
- ☐ Set shorter TTLs for content that changes frequently.
- ☐ Use purge/invalidation for freshness rather than universally short TTLs.
- ☐ Avoid sending no-cache/no-store on pages that should be public-cacheable.
- ☐ Verify generated headers from WordPress, plugins, CDN, and server rules.
- ☐ Ensure invalidation happens when content changes.
- ☐ Avoid purging the whole site for small content updates when targeted purges are possible.

Header review:

```
curl -I https://example.com/page/
```

Look for:

- ☐ Cache-Control
 - ☐ Expires
 - ☐ Vary
 - ☐ Set-Cookie
 - ☐ CDN cache status headers
 - ☐ custom platform cache headers
-

6. Application optimization

If a request reaches WordPress, optimize the application path.

- ☐ Identify expensive hooks, filters, and actions.
- ☐ Confirm callbacks run only where needed.
- ☐ Avoid global work during every request.
- ☐ Avoid excessive template complexity.
- ☐ Avoid template over-use where many partials perform redundant work.
- ☐ Cache expensive partial output when safe.
- ☐ Eliminate uncached functions on hot paths.
- ☐ Avoid remote HTTP calls during page rendering.
- ☐ Move non-critical work to async/background processing.

Hook review checklist:

- ☐ Does this callback run on frontend, admin, REST, AJAX, or all contexts?
 - ☐ Does it execute database queries?
 - ☐ Does it call external services?
 - ☐ Does it build data that could be cached?
 - ☐ Does it run for users/pages that do not need it?
-

7. Partial output caching

Use partial output caching for expensive fragments that cannot be solved by full-page cache alone. See `DEVELOPER_REFERENCE.md` §18 for the canonical cache-key safety checklist.

Cache-key safety checklist for fragment / partial-output caching

Before caching a fragment with `wp_cache_set()` or `set_transient()`, verify that the key includes every dimension that varies the output:

- Site / blog scope: `get_current_blog_id()` on multisite.
- Locale: `get_locale()` or `determine_locale()` if locale affects content.
- Currency / region: any active currency, region, or store context.
- Auth state: logged-in vs logged-out as separate keys.
- User identity: only when the fragment is intentionally per-user and you have evaluated privacy and cardinality implications.
- Role / capability: if the fragment differs by role or capability, include the relevant subset.
- A/B / personalization bucket: include the bucket identifier.
- Query parameters that change output: explicit allowlist, never raw `$_GET`.

Privacy and cardinality:

- Never cache personalized output under a shared key. One shared-key mistake can leak one user's content to another.
- Per-user keys grow cache footprint linearly with active users. Bound this with short TTLs and use them only when regenerate cost dominates.
- On persistent object caches, monitor key cardinality and memory pressure for any per-user fragment cache group.

Invalidation triggers:

- Source content change: `save_post`, custom post type hooks, ACF/CMB2 update hooks.
- User profile / role / capability change: `set_user_role`, `profile_update`.
- Locale / settings change.
- Multisite blog switch / domain change.
- [] Identify repeated expensive fragments: sidebars, related posts, navigation, widgets, API-derived blocks.
- [] Create cache keys that include only the context that changes output.
- [] Set an expiration appropriate to freshness needs.
- [] Invalidate when source data changes where necessary.
- [] Avoid caching personalized fragments under shared keys.

Example pattern:

```
<?php
$role_fragment = is_user_logged_in() ? 'logged_in' : 'logged_out';
$cache_key      = sprintf(
    'recent_posts_sidebar_v2_blog%d_locale%s_state%s',
    get_current_blog_id(),
```

```
        sanitize_key( determine_locale() ),
        $role_fragment
    );

$recent_posts_html = wp_cache_get( $cache_key, 'page_fragments' );

if ( false === $recent_posts_html ) {
    ob_start();
    // Generate the expensive sidebar output here.
    get_template_part( 'partials/recent-posts-sidebar' );
    $recent_posts_html = ob_get_clean();

    wp_cache_set( $cache_key, $recent_posts_html, 'page_fragments', HOUR_IN_SECONDS );
}

echo $recent_posts_html;
```

8. MySQL query optimization

Slow database work is one of the most common origin-side bottlenecks.

- [] Count total queries for the request.
- [] Identify slow queries.
- [] Identify duplicate queries.
- [] Identify unbounded queries.
- [] Identify queries that scan too many rows.
- [] Avoid expensive meta queries on high-cardinality data.
- [] Avoid taxonomy over-use for query patterns that do not scale.
- [] Avoid NOT IN when it forces inefficient exclusion scans.
- [] Avoid leading-wildcard LIKE searches on large tables.
- [] Use search infrastructure such as Elasticsearch/OpenSearch for heavy search workloads.
- [] Use EXPLAIN to understand execution plans.

WP_Query baseline example:

```
$query = new WP_Query(
    array(
        'post_type'      => 'post',
        'posts_per_page' => 5,
```

```
        'orderby'      => 'date',
        'order'        => 'DESC',
    )
);
```

Query anti-pattern checks:

- [] `posts_per_page` => -1 on large datasets.
 - [] missing pagination.
 - [] meta queries used as primary filters at scale.
 - [] taxonomy queries stacked into complex intersections/exclusions.
 - [] `post__not_in` or SQL `NOT IN` against large sets.
 - [] `LIKE '%term%'` on large text columns.
 - [] ordering by unindexed or computed values.
-

9. Unbounded queries

Unbounded queries are dangerous because they look safe during development and fail at enterprise scale.

- [] Search code for `posts_per_page` => -1.
- [] Search code for `numberposts` => -1.
- [] Search code for falsy IDs accidentally becoming broad queries.
- [] Add explicit limits.
- [] Paginate large result sets.
- [] Use caching for expensive repeated query results.
- [] Validate IDs and input before building queries.

Examples:

```
$post_id = intval( $my_unexpectedly_falsy_post_id );
$query_args = array(
    'p'          => $post_id,
    'posts_per_page' => -1,
);
$args = array(
    'post_type' => 'post',
    'date_query' => array(
        'after'  => '2021-01-01',
        'before' => '2021-12-31',
    ),
);
```

```
    ),  
);  
  
$query = new WP_Query( $args );
```

Safer pattern:

```
function limit_news_scope_query_filter( $query ) {  
    if ( is_admin() || ! $query->is_main_query() ) {  
        return;  
    }  
  
    if ( $query->is_category( 6 ) ) {  
        $query->set( 'posts_per_page', 20 );  
    }  
}  
add_action( 'pre_get_posts', 'limit_news_scope_query_filter' );
```

10. Taxonomies, exclusions, post meta, and LIKE

Taxonomies

- [] Use taxonomies for classification, not arbitrary high-cardinality data.
- [] Avoid overly complex taxonomy intersections/exclusions.
- [] Keep term relationships manageable.
- [] Use dedicated indexes/search systems for large faceted search use cases.

Example scope limiter:

```
function limit_news_scope_query_filter( $query ) {  
    if ( is_admin() || ! $query->is_main_query() ) {  
        return;  
    }  
  
    if ( $query->is_category() ) {  
        $query->set( 'posts_per_page', 20 );  
    }  
}  
add_action( 'pre_get_posts', 'limit_news_scope_query_filter' );
```

NOT IN

- ☐ Avoid large NOT IN exclusions.
- ☐ Prefer positive selection rules where possible.
- ☐ Cache exclusion sets if they must be computed.
- ☐ Test with production-scale data.

Example to scrutinize:

```
SELECT * FROM wp_posts
WHERE post_type = 'post'
      AND post_status = 'publish'
      AND ID NOT IN (1, 2, 3);
```

Post meta

- ☐ Do not treat post meta as a general-purpose relational store at scale.
- ☐ Avoid sorting/filtering large result sets by arbitrary meta values.
- ☐ Consider custom tables or search infrastructure for high-volume structured data.

LIKE

- ☐ Avoid leading wildcard searches on large columns.
- ☐ Prefer indexed prefix searches if acceptable.
- ☐ Use search infrastructure for real search workloads.

Examples:

```
SELECT * FROM wp_posts WHERE post_title LIKE '%WordPress%';
SELECT * FROM wp_posts WHERE post_title LIKE 'WordPress%';
```

The second pattern can be more index-friendly than the first, but still needs production-scale testing.

11. EXPLAIN-driven query review

Use EXPLAIN to verify how MySQL plans to execute a query.

- ☐ Run EXPLAIN on slow queries.

- ☐ Check which table is scanned.
- ☐ Check access type.
- ☐ Check possible keys.
- ☐ Check the key actually used.
- ☐ Check estimated rows scanned.
- ☐ Look for filesort or temporary table usage.
- ☐ Compare plans before and after query/index changes.

Examples:

```
EXPLAIN SELECT * FROM wp_posts WHERE post_date > '2023-01-01';
EXPLAIN
SELECT wp_posts.*, wp_postmeta.meta_value
FROM wp_posts
JOIN wp_postmeta ON wp_posts.ID = wp_postmeta.post_id
WHERE wp_posts.post_date > '2023-01-01';
```

12. Object caching

Cache-layer terminology

Three distinct WordPress caching subsystems are often conflated. `WP_CACHE` is a constant that allows WordPress to load the `wp-content/advanced-cache.php` drop-in for page caching (saving complete rendered HTML). `wp-content/object-cache.php` is a separate drop-in for persistent object caching (Redis/Memcached, caching database/query/application objects across requests). The Transients API uses object cache when persistent caching is configured, otherwise falls back to the `wp_options` table. See [REFERENCE-WP-Transients-Persistent-Object-Cache.md](#) and [DEVELOPER_REFERENCE.md](#) §4 for full treatment.

Object caching reduces repeated computation and repeated database access.

- ☐ Confirm a persistent object cache exists when the site needs one.
- ☐ Confirm cache service health.
- ☐ Track hit rate and miss rate.
- ☐ Cache expensive query results and computed values.
- ☐ Use cache groups intentionally.
- ☐ Keep keys stable and specific.
- ☐ Avoid excessive cache calls in tight loops.
- ☐ Avoid storing huge objects that increase memory pressure.
- ☐ Avoid caching personalized/private data with shared keys.

13. Transients and in-memory caching

Transients:

- ☐ Use transients for data that can expire.
- ☐ Set realistic expiration times.
- ☐ Account for either database-backed or object-cache-backed storage.
- ☐ Avoid relying on transients for data that must always exist.
- ☐ Clean up transient buildup when needed.

In-memory request-level caching:

- ☐ Use static variables for repeat computations within one request.
- ☐ Do not use request-local cache as a substitute for persistent cache across requests.

Example:

```
function getExpensiveComputationResult( $input ) {
    static $cache = array();

    if ( isset( $cache[ $input ] ) ) {
        return $cache[ $input ];
    }

    $result = perform_expensive_computation( $input );
    $cache[ $input ] = $result;

    return $result;
}
```

14. Race conditions and cache stampedes

Race conditions occur when multiple requests try to rebuild the same cache item at once. Cache stampedes happen when many requests miss simultaneously and all perform the expensive work.

- ☐ Add locking around expensive cache regeneration.
- ☐ Use soft TTL/hard TTL patterns where appropriate.
- ☐ Refresh important cache entries before they expire during high traffic.

- ☐ Avoid synchronized expiration for many hot keys.
- ☐ Add jitter to expirations where useful.
- ☐ Serve stale data briefly when freshness requirements allow it.

Lock example:

These lock patterns assume a persistent object cache with atomic add semantics, such as Redis or Memcached via `wp-content/object-cache.php`. On default WordPress with no persistent object cache, `wp_cache_add()` is request-local: every concurrent request can “acquire” the lock simultaneously, so it provides no cross-request coordination. Confirm a persistent object cache is in place before relying on this pattern.

Fallbacks when persistent object caching is not available:

- Use a database-backed mutex or a short-lived transient/option lock, accepting the database-write cost and failure modes.
- Use a platform-provided distributed lock service if the host offers one.
- Use stale-while-revalidate: serve the previous value while a warmer regenerates asynchronously.
- Pre-warm hot keys outside user requests with cron, deploy hooks, or scheduled actions.
- Restructure so the expensive operation never runs inside a user request.

```
$lock_key = 'my_cache_lock';
$lock_acquired = wp_cache_add( $lock_key, true, 'locks', 30 );

if ( $lock_acquired ) {
    $data = regenerate_expensive_cache_value();
    wp_cache_set( 'my_cache_key', $data, 'my_group', HOUR_IN_SECONDS );
    wp_cache_delete( $lock_key, 'locks' );
} else {
    $data = wp_cache_get( 'my_cache_key', 'my_group' );
}
```

15. WP-Cron and scheduled work

WP-Cron can create request-time spikes if scheduled work runs during user traffic.

- ☐ Identify due cron events.
- ☐ Identify slow recurring jobs.
- ☐ Run cron through real server cron where appropriate.
- ☐ Disable request-triggered WP-Cron only after the external runner is verified.

- ☐ Avoid heavy jobs during peak traffic.
- ☐ Batch large jobs.
- ☐ Ensure jobs are idempotent.
- ☐ Monitor failures and backlog.

Sequence matters. Install and verify the external runner before setting `DISABLE_WP_CRON`. If the constant is set first, scheduled events and Action Scheduler queues can stop until the runner works.

WP-CLI cron runner:

```
* * * * * cd /path/to/site && wp cron event run --due-now --quiet
```

Verify before changing `wp-config.php`:

```
wp cron event list --due-now
# Check scheduler logs and application logs for successful invocations.
```

Only after verification:

```
define( 'DISABLE_WP_CRON', true );
```

Fallback PHP runner:

```
* * * * * php /path/to/wordpress/wp-cron.php > /dev/null 2>&1
```

HTTP fallback:

```
* * * * * curl -s https://example.com/wp-cron.php?doing_wp_cron >/dev/null 2>&1
```

After setting the constant, confirm scheduled events continue to clear. Roll back by commenting out the constant and clearing OPcache if the backlog grows.

16. AJAX, REST, sitemaps, redirects, and external requests

AJAX and REST

- ☐ Identify `admin-ajax.php` calls on public pages.
- ☐ Identify REST API calls after initial render.
- ☐ Remove unnecessary polling.
- ☐ Cache public endpoint responses where safe.
- ☐ Move heavy endpoint work out of synchronous user requests.
- ☐ Authenticate private endpoints correctly.

Sitemaps

- ☐ Ensure sitemap generation does not perform expensive unbounded queries.
- ☐ Cache sitemap output when possible.
- ☐ Paginate large sitemaps.
- ☐ Avoid regenerating large sitemap structures on every request.

External requests

- ☐ Identify remote HTTP calls during page generation.
 - ☐ Add strict timeouts.
 - ☐ Cache remote responses.
 - ☐ Fail gracefully when the remote service is slow.
 - ☐ Move non-critical requests to background processing.
-

17. Client-side JavaScript and browser rendering

Even when backend performance is good, the browser can still feel slow.

- ☐ Identify heavy frontend scripts.
 - ☐ Remove duplicate libraries.
 - ☐ Avoid loading plugin scripts globally when only one page needs them.
 - ☐ Defer non-critical JavaScript.
 - ☐ Reduce long main-thread tasks.
 - ☐ Measure interaction delay after script changes.
 - ☐ Watch for AJAX calls triggered by frontend scripts after load.
-

18. Traffic patterns and high-traffic events

High-traffic events require preparation, not just reaction.

- ☐ Identify expected traffic sources and timing.
- ☐ Estimate peak requests per minute/second.
- ☐ Identify pages that will receive traffic.
- ☐ Pre-warm caches.
- ☐ Freeze risky deployments before the event.
- ☐ Audit cacheability of campaign landing pages.

- ☐ Remove avoidable query-string cache fragmentation.
- ☐ Confirm origin can survive expected cache misses.
- ☐ Confirm monitoring and alerting are active.
- ☐ Prepare rollback and incident contacts.

High-traffic preflight:

- ☐ CDN/edge cache hit ratio acceptable.
 - ☐ Origin TTFB acceptable under warm cache.
 - ☐ No heavy uncached widgets on landing page.
 - ☐ No slow external dependency in critical path.
 - ☐ No scheduled jobs during peak window.
 - ☐ Load test completed in a safe environment.
-

19. Load testing

Load testing should be coordinated and scoped; never surprise production.

- ☐ Get explicit approval.
- ☐ Define target URLs and traffic model.
- ☐ Define acceptable limits and stop conditions.
- ☐ Test cacheable and uncached paths separately.
- ☐ Monitor origin, database, object cache, and CDN during the test.
- ☐ Compare behavior to expected real traffic.

Example:

```
k6 run --vus 100 --duration 5m script.js
```

Record:

- ☐ requests per second
 - ☐ p50/p95/p99 latency
 - ☐ error rate
 - ☐ cache hit ratio
 - ☐ PHP/application time
 - ☐ database pressure
 - ☐ object-cache pressure
-

20. Measuring time spent

Use multiple tools because each sees a different layer.

PHP logging

- ☐ Use targeted timing around suspected code paths.
- ☐ Remove noisy debug logging after investigation.
- ☐ Do not expose timing details publicly.

Example:

```
function my_custom_function() {
    $start_time = microtime( true );

    // Code to be measured here.

    $end_time = microtime( true );
    $execution_time = $end_time - $start_time;

    error_log( 'my_custom_function took ' . $execution_time . ' seconds' );
}
```

New Relic

- ☐ Use APM traces to find slow transactions.
- ☐ Group by transaction type: page, admin, REST, AJAX, cron.
- ☐ Inspect database, external service, and PHP segment timing.

Example:

```
// Guard so the code is safe whether the New Relic PHP agent is loaded or not.
if ( extension_loaded( 'newrelic' ) && function_exists( 'newrelic_name_transaction' ) ) {
    // Name the current transaction so it appears as a distinct entry
    // in the New Relic APM transaction list. Avoid per-user or per-
    // URL names.
    newrelic_name_transaction( 'myplugin/expensive-path' );
}

// Optional: add custom attributes for filtering and querying in NRQL.
if ( function_exists( 'newrelic_add_custom_parameter' ) ) {
```

```
newrelic_add_custom_parameter( 'plugin_context', 'myplugin_expensive_path' );
}
```

Use `newrelic_name_transaction()` to give the current request a stable, low-cardinality name in APM. The New Relic transaction-start API takes a New Relic application name, not a transaction name, and is reserved for advanced transaction-lifecycle patterns such as queue workers manually ending one transaction and starting another.

Query Monitor

- ☐ Inspect query count and slow queries.
- ☐ Inspect hooks and component timing.
- ☐ Inspect HTTP API calls.
- ☐ Use custom timers for targeted sections.

Example:

```
do_action( 'qm/start', 'mytimer' );

foreach ( range( 1, 10 ) as $i ) {
    function_to_measure_time_spent( $i );
}

do_action( 'qm/stop', 'mytimer' );
```

21. Performance alerting

- ☐ Alert on high TTFB.
- ☐ Alert on elevated error rate.
- ☐ Alert on database saturation or slow queries.
- ☐ Alert on object-cache service failures.
- ☐ Alert on CDN cache hit ratio drops.
- ☐ Alert on queue/cron backlogs.
- ☐ Alert on external dependency latency.
- ☐ Alert on sudden traffic spikes.

Good alerts should include:

- ☐ affected URL or transaction

- ☐ severity
 - ☐ current value vs threshold
 - ☐ duration
 - ☐ likely owner
 - ☐ first diagnostic link
 - ☐ rollback/escalation instructions
-

22. Verification and definition of done

After each optimization:

- ☐ Re-run the same measurement from the baseline.
- ☐ Confirm the same user state and URL were tested.
- ☐ Confirm cache status is expected.
- ☐ Confirm no private data is cached publicly.
- ☐ Confirm no critical functionality regressed.
- ☐ Confirm no new PHP errors.
- ☐ Confirm query count/time improved if database was targeted.
- ☐ Confirm cache hit ratio improved if caching was targeted.
- ☐ Document before/after values.
- ☐ Document remaining bottlenecks.

Done means:

- ☐ The bottleneck has been identified, not guessed.
 - ☐ The fix addresses that bottleneck directly.
 - ☐ The improvement is measurable.
 - ☐ The change is safe under expected traffic.
 - ☐ Monitoring can detect regression.
-

Appendix A: Modern additions (verified against WordPress 6.9.4, 2026-05-18)

The WP VIP Learn Enterprise WordPress Performance course was recorded before several WordPress Core performance changes that matter at platform scale. This appendix covers them in enterprise terms. The unified developer reference at [DEVELOPER_REFERENCE.md](#) has fuller treatment.

Options API expansion (WordPress 6.4 and 6.6)

WordPress 6.4 added `wp_set_option_autoload()` and `wp_set_options_autoload()` for managing autoload state without touching the database directly. WordPress 6.6 expanded the autoload column to five values: 'on', 'off', 'auto', 'auto-on', 'auto-off'. Legacy 'yes'/'no' rows persist on upgraded sites.

Enterprise / platform implications:

- [] Any saved monitoring queries, dashboards, or platform health checks filtering only legacy 'yes' autoload rows need to switch to autoload IN ('yes', 'on', 'auto', 'auto-on'). This is a silent miss otherwise — newly written 6.6+ options stop showing up.
- [] WP 6.6 automatically marks options larger than ~150 KB as 'auto-off'. Custom platform code that writes large options with explicit autoload should respect this signal rather than fight it.
- [] Site Health adds a check labeled “Autoloaded options could affect performance” at ~800 KB total autoload size. Surface this in platform telemetry alongside alloptions size.
- [] On alloptions cache stampedes (the classic VIP issue), the WP 6.6 auto-skip heuristics reduce — but do not eliminate — the risk. Continue to set explicit autoload => false for known-large platform options.

```
// Platform code: prefer explicit autoload over relying on heuristics.  
add_option( 'my_platform_large_option', $value, '', false );
```

```
// To flip an existing option:  
wp_set_option_autoload( 'my_platform_large_option', false );
```

Speculation Rules / Speculative Loading (WordPress 6.8)

WordPress 6.8 (April 2025) merged speculative-loading support into Core via the [Speculation Rules API](#). Core’s effective default is prefetch with conservative eagerness, enabled on frontend requests only when pretty permalinks are on and the user is logged out. Core does not ship a dedicated settings UI for speculation rules in 6.8; customization happens through filters such as `wp_speculation_rules_configuration` and `wp_load_speculation_rules`, plus block-level CSS classes like `no-prefetch` and `no-prerender`.

The standalone [Speculative Loading plugin](#) extends Core with a plugin-provided admin screen and more aggressive modes, including prerender. When you opt into prerender, audit analytics SDKs for prerender-aware behavior and confirm that prerendered visits do not double-count or fire third-party scripts on hidden documents.

Enterprise considerations: edge cache hit ratio can rise on sites with predictable navigation patterns because prefetches may hit cache before the eventual navigation. Confirm session-keyed and personalized pages are excluded from speculative loading with Core filters, no-prefetch / no-prerender classes, or plugin-provided exclusion controls. Do not use older data-prefetch="false" wording as the recommended Core opt-out mechanism.

WordPress 6.9 performance deltas

WordPress 6.9 shipped in November 2025; WordPress 6.9.4 was the current patch release verified on 2026-05-18. The most material 6.9 performance changes for this guide are:

- Frontend loading: the [WordPress 6.9 Frontend Performance Field Guide](#) covers script fetchpriority, footer script-module printing, emoji script-module changes, block stylesheet handling, hidden-block asset omission, template enhancement output buffering, RSS feed caching, video block CLS fixes, and related frontend work.
- WP-Cron spawn timing: Core now spawns WP-Cron at shutdown rather than earlier in the request lifecycle, reducing user-facing latency for request-triggered cron. Sites with a verified external cron runner remain the preferred high-traffic pattern.
- Query/cache changes: the [WordPress 6.9 Field Guide](#) should be checked before relying on plugin code that assumes older query-cache key behavior or intercepts Core caching internals.

Treat this as a delta inventory, not a substitute for measurement: verify impact on the specific site with the same baseline and rollback discipline used elsewhere in this guide.

fetchpriority on LCP image (WordPress 6.3+)

Core now sets fetchpriority="high" on the detected LCP image automatically. At enterprise scale this affects how the platform should advise on hero-image preload patterns:

- [] Audit any platform-level theme/plugin that emits <link rel="preload" as="image"> for hero images — it may now be redundant or, worse, may compete with Core's fetchpriority heuristic by preloading a different candidate.
- [] Confirm AMP / Block Theme combinations correctly identify the LCP candidate after 6.3.

Performance Lab plugin

[Performance Lab](#) is the Core team's feature-plugin discovery and management layer. As

features mature inside the Performance Lab ecosystem they either graduate into Core or remain as standalone plugins; the catalog rotates over time. Verify the current featured plugins on the Performance Lab plugin page before recommending any specific module.

Featured plugins as of 2026-05-18 include:

- Embed Optimizer — improves embeds that otherwise add third-party request and rendering cost.
- Enhanced Responsive Images — refines responsive image sizing decisions, including lazy-loaded images.
- Image Placeholders — formerly Dominant Color Images; uses a CSS background placeholder while an image loads.
- Image Prioritizer — automates `fetchpriority` and related loading decisions for likely LCP candidates.
- Instant Back/Forward — improves browser back/forward-cache behavior where applicable.
- Modern Image Formats — formerly named for WebP uploads; stores additional WebP/AVIF versions of uploaded images and serves modern formats to supporting browsers.
- Optimization Detective — opt-in real-user measurement that informs Core/Performance Lab decisions; dependency for some other featured plugins.
- Performant Translations, Speculative Loading, and View Transitions — additional featured plugins whose relevance depends on the site and rollout risk.

Recommendation: evaluate Performance Lab on staging, look at the current Featured Plugins list at the verification time, and adopt specific plugins by name only when they match the site's documented performance need.

Core Web Vitals: INP replaced FID (March 2024)

INP is now the responsiveness metric of record, replacing FID. Thresholds at the 75th percentile across real users:

Metric	Good	Needs	Poor
		improvement	
LCP	≤ 2.5 s	2.5–4.0 s	> 4.0 s
INP	≤ 200 ms	200–500 ms	> 500 ms
CLS	≤ 0.1	0.1–0.25	> 0.25

Enterprise platform reporting should retire FID dashboards and ensure RUM SDKs re-

port INP. INP is more sensitive to JavaScript main-thread work than FID was, so customer-success conversations about “suddenly worse mobile” often trace to long-running scripts that FID would not have penalized.

Editorial correction to source lesson

WP VIP Learn lesson 34 (WP Cron) shows the cron URL as `http://yourdomain.com/wp-cron.php?doing_wp_cron`. That is insecure for any modern production site — credentials, cookies, and the WP-Cron token would travel in cleartext. Use `https://` matching the site’s canonical scheme. (This operational checklist at §15 already uses HTTPS.)

References

- [Options API: Disabling autoload for large options \(Make WP Core\)](#)
- [New option functions in 6.4 \(Make WP Core\)](#)
- [WordPress 6.6 Performance Improvements](#)
- [Speculative Loading in 6.8](#)
- [WordPress 6.8 Performance Improvements](#)
- [wp_set_option_autoload\(\)](#)
- [Performance Lab plugin](#)
- [web.dev — Interaction to Next Paint](#)
- [Introducing INP to Core Web Vitals \(Google Search Central\)](#)